Insurance Claims & Fraud Analysis Dashboard

1. Project Overview

Project Title: Insurance Claims & Fraud Analysis Dashboard

Tool: Microsoft Power BI

Dataset: Insurance Claims Dataset

Records: 1,000 insurance claims

Features: 39 columns

Project Type: Business Intelligence / Data Analytics

Primary Objective: Analyze insurance claim patterns, claim amounts, policy characteristics, incident severity, and reported fraud to support data-driven insurance decision-making.

This project transforms a raw insurance claims dataset into an interactive Power BI dashboard. The dashboard combines KPI cards, trend analysis, category comparisons, demographic analysis, incident-level summaries, and slicers so users can explore claim performance and fraud patterns from multiple perspectives.

***Important:** This is a descriptive and diagnostic analytics project. It identifies patterns and relationships in the available data; it does not claim to predict future fraud or establish causation.*

---

2. Business Problem

Insurance companies process a large number of claims involving different policy types, customers, vehicles, incident categories, and claim amounts. Reviewing these claims individually can make it difficult to identify patterns in financial exposure and potentially fraudulent activity.

The business questions addressed by this project include:

• How much has been claimed overall?

• What is the average claim amount?

• What proportion of claims are reported as fraudulent?

• How do claim amounts vary over incident dates?

• Which policy coverage limits generate the highest claim amounts?

• How does incident severity relate to claim volume, claim value, and fraud?

• Which incident types contribute most to the total claim amount?

• How do claim patterns differ across age groups and gender?

• Which incident categories deserve closer investigation?

---

3. Objectives

Primary Objectives

• Analyze the overall insurance claims portfolio.

• Measure total claim value and average claim value.

• Monitor the proportion of reported fraudulent claims.

• Compare claim activity across policy coverage limits.

• Analyze claim trends by incident date.

• Understand the relationship between incident severity and claim outcomes.

• Compare claim behavior across incident types.

• Explore demographic patterns using age groups and gender.

• Provide interactive filtering for fraud status, policy CSL, and incident date.

• Present business insights in a simple dashboard suitable for decision-makers.

---

4. Dataset Description

The dataset contains **1,000 insurance claim records and 39 variables**.

Major Data Categories

Category	Example Fields	Purpose
Customer / Policy	age, months_as_customer, policy_state, policy_csl, policy_deductable	Understand customer and policy characteristics
Premium / Financial	policy_annual_premium, umbrella_limit, capital-gains, capital-loss	Analyze financial exposure
Incident	incident_date, incident_type, incident_severity, incident_state, incident_city	Analyze claim circumstances
Damage / Claim	total_claim_amount, injury_claim, property_claim, vehicle_claim	Measure claim financial impact
Investigation	authorities_contacted, witnesses, police_report_available	Examine investigation-related attributes
Vehicle	auto_make, auto_model, auto_year	Analyze vehicle-related patterns
Fraud	fraud_reported	Identify reported fraud cases

Target / Business Indicator

`fraud_reported` is the main fraud indicator:

• Y = reported as fraudulent

• N = not reported as fraudulent

---

5. Data Quality Assessment

The dataset contains complete values for most fields. However, several categorical fields contain placeholder values represented by `?`, and `authorities_contacted` contains missing values.

Observed data-quality issues include:

• `authorities_contacted`: 91 missing values.

• `property_damage`: 360 records contain `?`.

• `police_report_available`: 343 records contain `?`.

• `collision_type`: 178 records contain `?`.

These values should be treated as **unknown / unavailable**, rather than automatically interpreted as a meaningful business category.

The dataset also contains date fields stored as text and categorical fields requiring appropriate categorical handling in Power BI.

---

6. Data Preparation

The dashboard preparation process follows a structured BI workflow:

Step 1 — Data Import

The insurance CSV dataset was imported into Power BI.

Step 2 — Data Type Validation

Data types were reviewed for:

• Numeric fields such as claim amounts, premiums, age, witnesses, and vehicle counts.

• Date fields such as `policy_bind_date` and `incident_date`.

• Categorical fields such as policy state, incident type, severity, gender, and fraud status.

Step 3 — Missing / Unknown Value Review

Missing values and ? placeholders were identified so that they would not be incorrectly interpreted during analysis.

Step 4 — Analytical Fields

The dashboard uses business-friendly analytical fields/measures such as:

• Claim Count

• Total Claim Amount

• Average Claim Amount

• Fraud Claim %

• Age Group

Step 5 — Dashboard Validation

Dashboard measures were cross-checked against the underlying dataset to ensure that the headline KPIs represented the 1,000 source records correctly.

---

7. Key KPI Results

Based on the supplied dataset:

KPI	Result
Total Claims	**1,000**
Total Claim Amount	**52,761,940**
Average Claim Amount	**52,761.94**
Reported Fraudulent Claims	**247**

KPI	Result
Fraud Claim Rate	**24.7%**
Median Claim Amount	**58,055**

The reported fraudulent claims represent **24.7% of all claims**. Although they account for 24.7% of claim records, their associated claim amount is approximately **14.89 million**, or **28.2% of the total claim amount**. This indicates that reported fraudulent claims have a higher average claim amount than non-fraudulent claims in this dataset.

• Average fraudulent claim: **60,302.11**

• Average non-fraudulent claim: **50,288.61**

---

8. Power BI Dashboard Design

The supplied Power BI report contains a single dashboard page with interactive visuals.

KPI Cards

The dashboard includes KPI cards for:

• **Total Claims**

• **Total Claim Amount**

• **Average Claim Amount**

• **Fraud Claim %**

These provide an immediate portfolio-level summary.

Claim Trend Visual

A combined line and clustered-column visual analyzes:

• Incident Date

• Total Claim Amount

• Claim Count

This helps compare the volume and financial value of claims across incident dates.

Policy CSL Analysis

A clustered bar chart compares policy Coverage Single Limit (CSL) categories using:

• Total Claim Amount

• Claim Count

• Fraud Claim %

This allows users to understand whether claim exposure differs by policy coverage level.

Incident Severity Analysis

A donut chart summarizes claim count by incident severity, with claim value and fraud percentage available as supporting metrics.

Demographic Analysis

A clustered column chart compares claim values across **Age Group** and **insured sex**, with claim count and fraud percentage available for deeper analysis.

Incident Type Summary

A matrix/pivot-style visual provides:

• Claim Count

• Total Claim Amount

• Fraud Claim %

by incident type.

Interactive Slicers

The dashboard includes slicers for:

• Fraud Reported

• Policy CSL

• Incident Date

These allow users to dynamically filter the dashboard and investigate specific segments.

---

9. Exploratory Findings

9.1 Incident Severity

Incident severity shows a strong relationship with both claim value and reported fraud.

Incident Severity	Claims	Total Claim Amount	Fraud Rate
Major Damage	276	17,682,540	**60.5%**
Total Loss	280	17,382,740	12.9%
Minor Damage	354	17,219,510	10.7%
Trivial Damage	90	477,150	6.7%

Insight: Major Damage claims have the highest reported fraud rate at **60.5%** and the highest average claim amount among the severity categories. This makes major-damage claims a particularly important segment for investigation and control review.

9.2 Incident Type

Incident Type	Claims	Total Claim Amount	Fraud Rate
Single Vehicle Collision	403	25,971,520	29.0%
Multi-vehicle Collision	419	25,825,910	27.2%
Vehicle Theft	94	518,620	8.5%
Parked Car	84	445,890	9.5%

Insight: Collision-related incidents dominate the financial exposure. Single-vehicle collisions generate the highest total claim amount, while multi-vehicle collisions have the highest claim count.

9.3 Policy CSL

Policy CSL	Claims	Total Claim Amount	Fraud Rate
100/300	349	19,287,730	25.8%
250/500	351	17,954,960	26.2%
500/1000	300	15,519,250	21.7%

Insight: The 100/300 CSL segment contributes the largest total claim amount, while the 250/500 segment has the highest reported fraud rate among the three CSL categories.

9.4 Incident State

The highest total claim amount occurs in **NY**, with approximately **14.77 million** across 262 claims.

The highest observed reported fraud rate among incident states is **OH at 43.5%**, although the OH segment contains only 23 claims. This illustrates why fraud-rate comparisons should always be interpreted together with claim volume.

9.5 Gender Comparison

Gender	Claims	Total Claim Amount	Fraud Rate
Female	537	28,645,360	23.5%
Male	463	24,116,580	26.1%

Insight: Female customers account for more claims and higher total claim value because they represent a larger share of the records. Male customers show a slightly higher reported fraud rate in this dataset.

This should be treated as a descriptive observation rather than evidence that gender causes fraudulent behavior.

---

10. Claim Amount Composition

The dataset records three major claim components:

Claim Component	Total
Vehicle Claim	**37,928,950**
Injury Claim	**7,433,420**
Property Claim	**7,399,570**

Vehicle claims are the largest component of the claim portfolio.

The component totals add up to the overall total claim amount:

37,928,950 + 7,433,420 + 7,399,570 = 52,761,940

This provides an internal consistency check on the claim-value fields.

---

11. Business Insights

The analysis produces several actionable observations:

Insight 1 — Fraud requires financial prioritization

Reported fraudulent claims represent 24.7% of claim records but approximately 28.2% of the total claim value. Fraud review can therefore consider both **claim frequency and financial exposure**.

Insight 2 — Major damage is a high-risk segment

Major Damage claims have a reported fraud rate of 60.5%, substantially above the overall 24.7% rate. This segment could receive stronger verification and investigation controls.

Insight 3 — Collision claims drive the portfolio

Single-vehicle and multi-vehicle collisions together account for the overwhelming majority of claim value. Collision claims should therefore receive significant attention in claims operations and risk monitoring.

Insight 4 — Vehicle damage is the main cost driver

Vehicle claims account for approximately 37.93 million of the 52.76 million total claim amount, making vehicle-related losses the largest financial component.

Insight 5 — High percentages need volume context

Some incident-state or category groups may show high fraud percentages with relatively few claims. Decision-makers should consider claim count alongside fraud rate before prioritizing a segment.

---

12. Dashboard Interactivity

The dashboard is designed for exploratory analysis rather than a static report.

Users can filter the report using:

• **Fraud Reported:** Compare fraudulent and non-fraudulent cases.

• **Policy CSL:** Investigate different coverage-limit segments.

• **Incident Date:** Focus on selected periods.

Because the visuals are connected, changing a slicer updates the relevant KPIs and charts, allowing users to perform quick drill-down analysis.

---

13. Technology Stack

• **Microsoft Power BI Desktop**

• **Power Query** for data preparation

• **DAX** for analytical measures

• **CSV** as the source dataset

• **GitHub** for project documentation and portfolio presentation

---

14. Skills Demonstrated

This project demonstrates practical skills in:

• Data cleaning and validation

• Data type management

• Power Query

• DAX measures

• KPI development

• Data visualization

• Interactive dashboard design

• Business-oriented data analysis

• Fraud pattern analysis

• Financial analysis

• Exploratory data analysis

• Data storytelling

• GitHub project documentation

---

15. Limitations

• The dataset contains only 1,000 records, so findings should not automatically be generalized to a real insurance portfolio.

• The dataset includes unknown/placeholder values represented by ?.

• The dashboard is descriptive and diagnostic; it is not a machine-learning fraud prediction system.

• Reported fraud is an observed label in the dataset and should not be interpreted as a definitive legal determination of fraud.

• Some groups have relatively small sample sizes, so their fraud percentages can fluctuate substantially.

• The dataset appears to represent a historical/static sample rather than a live claims environment.

---

16. Future Enhancements

The project can be extended with:

• A dedicated fraud-risk scoring model using machine learning.

• Drill-through pages for individual claim investigation.

• Additional time intelligence and monthly trend analysis.

• Geographic visualization of incident locations/states.

• Claim severity benchmarking.

• Automated anomaly detection.

• A claims investigation priority score.

• More advanced DAX calculations for year-over-year and segment-level comparisons.

• Deployment to Power BI Service with scheduled refresh when a live data source is available.

---

17. Recommended GitHub Repository Structure

```
Insurance-Claims-Fraud-Analysis/
■
■■■ README.md
■■■ Insurance_PowerBI.pbix
■■■ insurance_dataset.csv
■■■ report/
■   ■■■ Insurance_Project_Report.pdf
■■■ images/
    ■■■ dashboard.png
```

If the PBIX file is too large for GitHub's normal file workflow, use **Git LFS** or provide the file through an appropriate release/storage location.

---

18. Project Conclusion

The Insurance Claims & Fraud Analysis Dashboard converts raw insurance claim records into a decision-support dashboard focused on **claim volume, financial exposure, incident characteristics, policy coverage, and reported fraud**.

The analysis shows that the portfolio contains **1,000 claims worth 52.76 million**, with a **24.7% reported fraud rate**. Major Damage incidents stand out because of their high reported fraud rate, while collision incidents and vehicle claims represent the largest areas of financial exposure.

The project demonstrates how Power BI can be used not only to create attractive dashboards, but also to structure raw business data into measurable KPIs, identify meaningful patterns, and communicate findings in a form that can support operational review.

---

19. Portfolio Summary

Insurance Claims & Fraud Analysis Dashboard | Power BI

Built an interactive Power BI dashboard using 1,000 insurance claim records to analyze claim volume, financial exposure, incident severity, policy coverage, and reported fraud. Developed KPI measures for total claims, total claim amount, average claim amount, and fraud rate, and created interactive visuals and slicers to support segment-level analysis. Identified major-damage incidents and collision-related claims as key areas for financial and fraud-risk review.

---

20. Data Note

The numerical findings in this report were calculated from the supplied `insurance dataset - LMS` CSV and cross-checked against the structure of the supplied Power BI report. Values may change if the source dataset or Power BI calculations are subsequently modified.